Model B vs Model A

`MAM_modelB.py` and differences from `modified_Ginzburg_Landau_MAM.py`

Shih-Huan Huang

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Hamiltonian: Model A

(Original `modified_Ginzburg_Landau_MAM.py`)

$$H = \sum_{\text{boxes}} \left[\left(D\nabla^2\rho + \rho - \rho^3 + \kappa\langle\rho^2\rangle \right)\theta \; + \; \frac{a}{2}\,\theta^2 \right]. \tag{1}$$

- Laplacian: discrete 3-point stencil via `np.roll`.
- Nonlocal coupling $\kappa\langle\rho^2\rangle$; noise term $a\theta^2/2$ is **not** conserved.

Hamiltonian: Model B

(`MAM_modelB.py`)

$$H = \sum_{\text{boxes}} \left[\nabla^2 \left(-D\nabla^2\rho - \rho + \rho^3 \right) \theta \; - \; \frac{a}{2} \theta \, \nabla^2 \theta \right]. \tag{2}$$

- Deterministic part: $\nabla^2 \mu$ with $\mu = -D\nabla^2\rho - \rho + \rho^3$ (chemical potential).
- Noise part: $-\frac{a}{2}\theta\nabla^2\theta$ is **conserved** (gradient form).
- No κ; no nonlocal coupling.
- Laplacian(∇^2) implemented in **Fourier space**: `lap(field)` = IFFT($-k^2$·FFT(field)); uses `k2`, and `k4` for biharmonic terms (∇^4) where needed.

Laplacian: Model A vs Model B

Model A: real-space finite difference

$$\nabla^2 f \to \frac{f_{i+1} + f_{i-1} - 2f_i}{h^2}, \quad \texttt{np.roll}. \tag{3}$$

Model B: Fourier space

$$\nabla^2 f \to \mathcal{F}^{-1}(-k^2\,\mathcal{F}(f)), \quad \texttt{k_vec},\ \texttt{k2} = k^2,\ \texttt{k4} = k^4. \tag{4}$$

`lap(field)` FFTs along the spatial axis (last axis), multiplies by $-k^2$, then IFFTs. Used for both ρ and θ in H and in $\partial H/\partial\rho$, $\partial H/\partial\theta$.

$\partial H/\partial\rho$ and $\partial H/\partial\theta$: Model B

In Model B the derivatives are (in the code):

$$\frac{\partial H}{\partial \rho} = -D\,\nabla^2(\nabla^2\theta) - (1-3\rho^2)\,\nabla^2\theta,$$
$$\frac{\partial H}{\partial \theta} = \nabla^2(-D\nabla^2\rho - \rho + \rho^3) - a\,\nabla^2\theta.$$

Implemented using `lap(rho)`, `lap(theta)`, and `lap(lap_rho)` etc.; no `np.roll`. Reaction terms `reaction_U` and `reaction_V` are then built as in Model A (same U, V split), then FFT in the copy direction and tridiagonal solve.

Tridiagonal solve: Model A vs Model B (matrix form)

In Fourier space, for each wavenumber k we solve a linear system in the **path (copy)** index j. Let $r = \Delta\tau/\Delta s$ and $\eta_k = \gamma\,\Delta\tau\,D\,k^4$ (IMEX scalar). **Example:** V-field (lower triangular B_k). **Model A:**

$$B_k\,\mathbf{V}_k = \mathbf{b}_k, \quad \mathbf{b}_k = \begin{bmatrix} \mathrm{BC}_0 \\ V_1^{\mathrm{old}} + \Delta\tau\,F_1 \\ \vdots \\ V_{N-1}^{\mathrm{old}} + \Delta\tau\,F_{N-1} \end{bmatrix}.$$

B_k is lower triangular (e.g. diagonal $1+r$, subdiagonal $-r$; first row encodes boundary). **Model B (IMEX):** Add $\eta_k I$ to the matrix and $\eta_k\,\mathbf{V}_k^{\mathrm{old}}$ to the RHS:

$$(B_k + \eta_k I)\,\mathbf{V}_k = \mathbf{b}_k + \eta_k\,\mathbf{V}_k^{\mathrm{old}}.$$

Diagonal of the solved matrix becomes $(1 + r + \eta_k)$ etc. Same `solve_triangular(B_..., ..., lower=True)`; only entries and RHS differ.

IMEX: what it does & why physics is unchanged

In Fourier space the biharmonic term is $\sim Dk^4$. Pure explicit stepping gives $U_{\mathrm{new}} \approx (1 - \Delta\tau Dk^4)U_{\mathrm{old}} \Rightarrow$ high-k blows up. IMEX puts $\eta_k U_{\mathrm{new}}$ on the **left** (diagonal) and $\eta_k U_{\mathrm{old}}$ on the **right**. Then $U_{\mathrm{new}} = \mathrm{RHS}/(1+\eta_k)$: high-$k$ is **damped**. γ (e.g. 2.0) controls the strength.

At **convergence**, $\mathbf{V}_k = \mathbf{V}_k^{\mathrm{old}}$, so the implicit term $\eta_k \mathbf{V}_k$ and the explicit term $\eta_k \mathbf{V}_k^{\mathrm{old}}$ in $(B_k + \eta_k I)\mathbf{V}_k = \mathbf{b}_k + \eta_k \mathbf{V}_k^{\mathrm{old}}$ cancel. The limit satisfies the true equation; the stabilizer only affects the *path* to the minimum, not the minimum itself.

Endpoints ρ_1, ρ_2: Model A vs Model B

Model A: Homogeneous endpoints. $\rho_1 = \rho_- \cdot \mathbf{1}$, $\rho_2 = \rho_+ \cdot \mathbf{1}$ (all boxes equal); $\rho_\pm$ from roots of cubic (with κ). `solRho = np.sort(np.roots([-1, kappa, 1, 0]))`.

Model B: Inhomogeneous endpoints (spatial profile). E.g. `rho1[0]=solRho[0]`, `rho1[L//2]=solRho[2]` ("left low, middle high") and `rho2` the other way; `solRho = [-1, 0, 1]` fixed. Path connects two different spatial patterns with the same total mass.

Mass conservation in Model B

Model B conserves total mass $\int \rho$. In the code, after each half-step (after updating U or V and recovering ρ, θ):

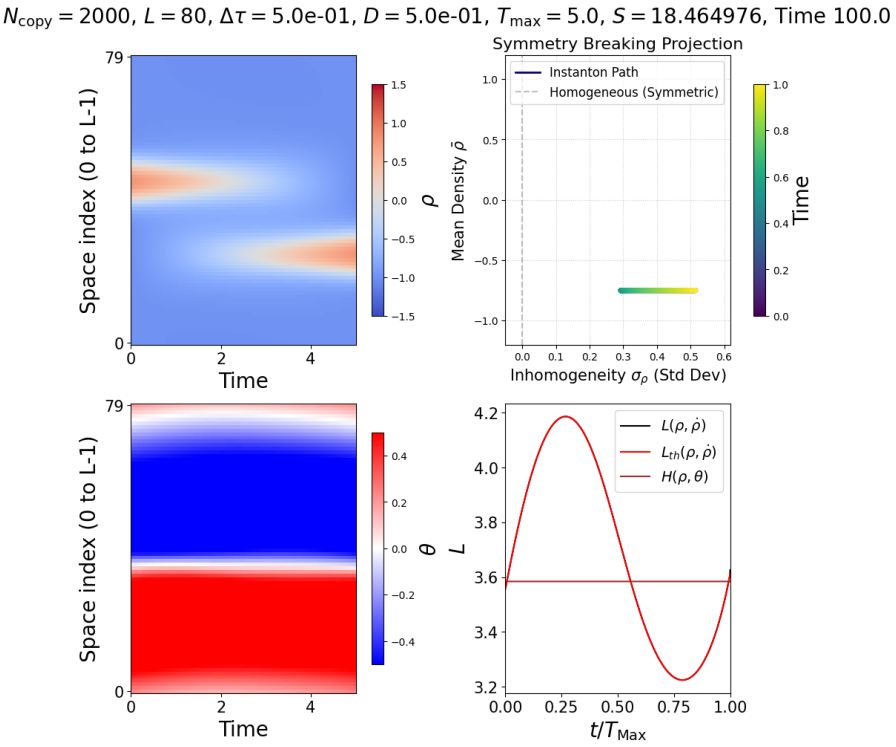
- `target_mass = np.mean(rho1)` (or rho1.real for consistency).
- `rho = rho - np.mean(rho, axis=1, keepdims=True) + target_mass`.
- `theta = theta - np.mean(theta, axis=1, keepdims=True)` (zero-mean θ).

Then re-enforce boundary: `rho[0,:]=rho1`, `rho[Ncopy-1,:]=rho2`. Initial path bump is also mean-subtracted: `bump = bump - np.mean(bump)`.

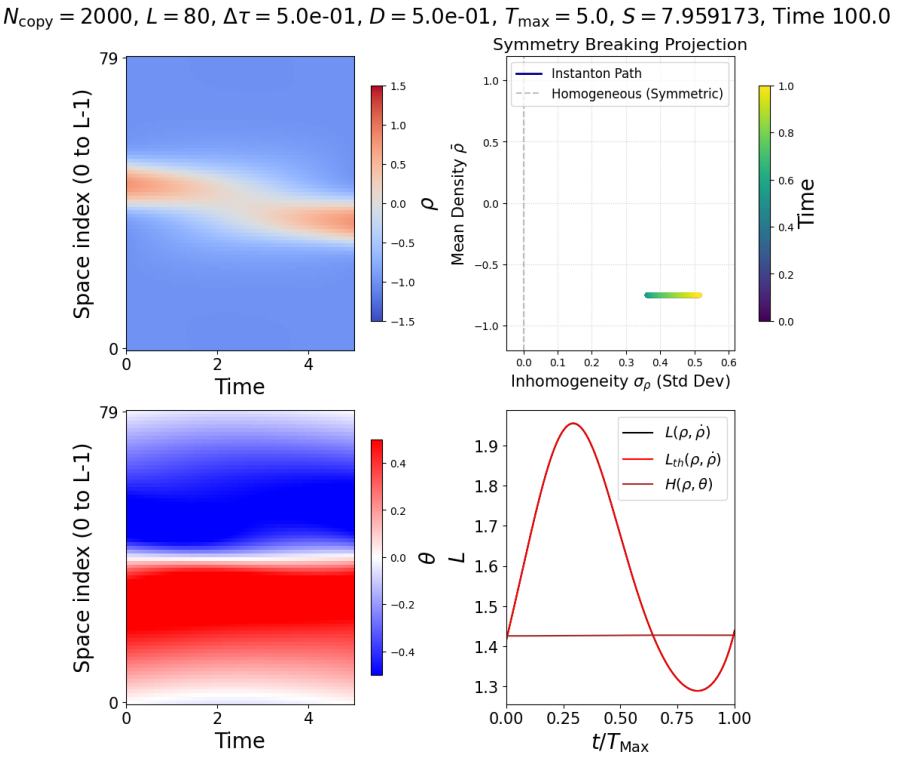
Parameters and extra output (Model B)

- **Parameters:** Model B uses `L=2`, `Ncopy=400`, `Tmax=2`, `iterations=2000`, `D=0.001`, no `kappa`; `solRho = [-1, 0, 1]`.
- **Lagrangian_Theo:** Model B defines `Lagrangian_Theo(h, rho, theta)` $= -0.5\, a\, \theta\, \nabla^2 \theta$ (theoretical Lagrangian density for conserved noise). Plot includes L_{th} alongside L and H.

Example: Model B instanton ($L = 80$, 200 iterations)



Run 1



Run 2

Summary: Model B vs Model A

- **Physics:** Model B = conserved dynamics (gradient form + conserved noise); Model A = non-conserved with κ and θ^2 noise.
- **Hamiltonian:** Model B uses $\nabla^2\mu\,\theta$ and $-\frac{a}{2}\theta\nabla^2\theta$; Model A uses $(D\nabla^2\rho + \rho - \rho^3 + \kappa\langle\rho^2\rangle)\theta + \frac{a}{2}\theta^2$.
- **Laplacian:** Model A = real-space stencil; Model B = FFT (`lap`, `k2`, `k4`).
- **Tridiagonal:** Model B adds IMEX stabilization (`gamma`, `k4`) to diagonal and RHS.
- **Endpoints & mass:** Model B uses inhomogeneous ρ_1, ρ_2 and enforces mass conservation each half-step; Model A uses homogeneous endpoints, no mass correction.